# CS-104 Quiz 1

Fall 2026

Name: _____

*This is an in-class, closed-book quiz. If you have any questions, please feel free to ask them. When you are finished, you can turn it in and go.*

**1. (4 points)**

Give the value of each Python expression below. If it raises an error instead, name the error (e.g., "IndexError").

```
>>> 'x' + 'y'
```
*'xy' — example answer*

```
>>> 15 // 4
```

```
>>> 3 ** 4
```

```
>>> [10, 20, 30][5]
```

```
>>> {'x': 1, 'y': 2}['z']
```

**2. (6 points)**

Write an algorithm (in plain English, not Python code) that prompts the user for a distance in miles and prints the equivalent distance in kilometers. Use the **Gather-Compute-Use** pattern from class, and note that `kilometers = miles * 1.60934`.

**3. (4 points)**

Write a code segment that inputs a number of total seconds from the user (as an integer) and prints how many whole minutes and remaining seconds that is. Here is an example run:

```
Total seconds: 130
Minutes: 2, remaining seconds: 10
```

Note that you should: - implement the prompt as shown (`Total seconds: `). - use `//` and `%` as appropriate to compute the minutes and remaining seconds. - print the result in the format shown above.

Just write a code segment; you don't need to include documentation, an algorithm, or test cases.

**4. (3 points)**

a) Write a statement that creates a list called `sizes` containing the strings `'small'`, `'medium'`, and `'large'`.

b) Write a statement that changes the **first** element of `sizes` to `'tiny'`.

**5. (3 points)**

Consider the dictionary defined below.

```
ages = {'Sam': 21, 'Lee': 19}
```

Write a statement that adds a new entry mapping `'Kim'` to `22`, or explain why this cannot be done.